(TR-104) PYTHON WITH AI/ML

Training Day 61 Report

Date: 6 April 2026

Topic Covered: Backend Workflow Engine, MCP Tool Integration & Frontend Flow Builder

The sixty-first day of training focused on building a complete agentic workflow system combining FastAPI backend, MCP tool integration, and a React-based visual workflow builder. The session emphasized end-to-end execution of node-based AI workflows and tool orchestration.

- Created FastAPI routes including /run-workflow, /tools, and /model for workflow execution and model interaction.
- Built a workflow execution engine supporting node-based processing architecture.
- Implemented multiple node types including inputNode, ollamaNode, agentNode, and outputNode.
- Designed workflow traversal logic using linear and DFS-based execution with loop detection.
- Integrated Ollama API (/api/chat) for generating model responses within workflows.
- Enabled tool calling support using structured tool parameters in model requests.
- Built an agentic execution loop for multi-step reasoning and decision-making workflows.
- Implemented execution limits using max turns to prevent infinite loops in agent workflows.
- Added dynamic tool selection mechanism based on user input and workflow configuration.
- Integrated error handling and structured logging for debugging workflow execution.
- Developed a custom MCP client supporting tools like calculator, Wikipedia search, Google search, and weather API.
- Designed a tool schema compatible with LLM tool-calling standards.
- Implemented async tool execution using httpx for non-blocking API calls.

- Created MCP Manager for handling multiple MCP servers dynamically.
- Supported addition and removal of external MCP servers with session management.
- Built React Flow-based frontend for drag-and-drop workflow creation.
- Implemented node types including Input, Ollama, Agent, and Output nodes in UI.
- Enabled edge-based node connections for workflow definition.
- Integrated backend execution trigger from frontend using Run Workflow feature.
- Displayed real-time workflow output inside output nodes for user interaction.
- Strengthened understanding of full-stack AI workflow systems combining backend orchestration, MCP tools, and frontend visualization.
- **GitHub Link:** https://github.com/JaspinderKaurWalia26/workflow_system